

Generative diffusion model with inverse renormalization group flows

Masuki, K., Ashida, Y. (2025). [arXiv:2501.09064](#).

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Diffusion models: two practical issues

Diffusion $\leftrightarrow$ RG: conceptual similarity

Physics $\leftrightarrow$ ML dictionary: establish a mapping

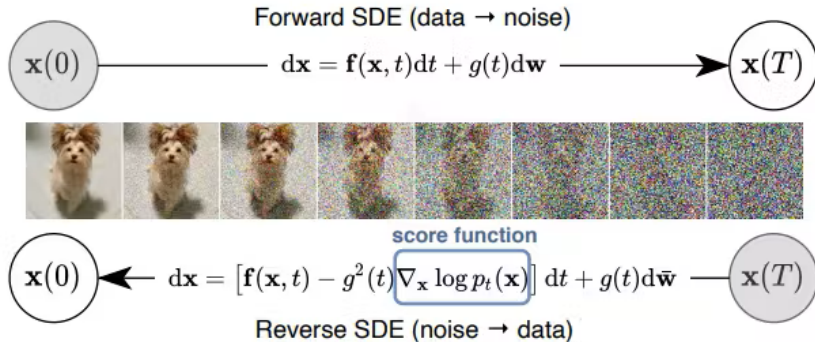
Exact RG

polchinski equation

Mapping to the Diffusion Model

Applications to image generation

conclusion



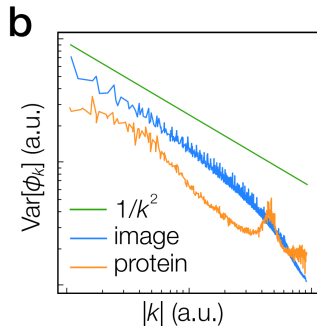
FP equation:

$$\partial_t p_t(x) = -\nabla \cdot (f(x, t) p_t(x)) + \frac{1}{2} \nabla \cdot (\nabla (g(t)^2 p_t(x))).$$

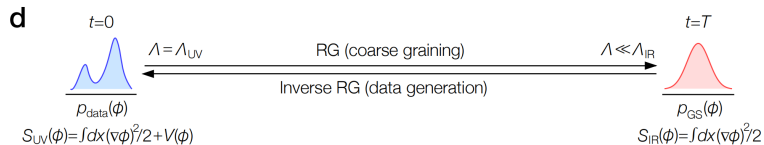
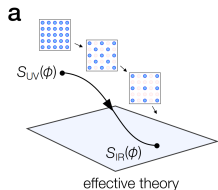
Discrete version:

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

- **Noise schedule is heuristic:** choose $\{\beta_t\}$ ($g(t)$) mostly by trial-and-error.
- **Ignores explicit multiscale:** natural images often show a power-law spectrum $S(k) \propto k^{-\alpha}$. Adding Gaussian noise will cause:
 1. *Forward:* high- k modes become nearly pure noise early, while low- k remains informative.
 2. *Reverse:* even if high- k modes are already drowned in noise at some t , the reverse model still must spend capacity trying to infer fine details from them.



Can we process different scale at different step?



Diffusion: a path of probability density $\{p_t(x)\}$ with $p_{t=0} = p_{data}$, $p_{t=T} \approx \mathcal{N}(0, I)$.
 $x \in \mathbb{R}^d$; p_t is a function on data space.

RG: a path of probability density $\{P_\Lambda[\phi]\}$ with $P_{\Lambda=\Lambda_{UV}} = P_{bare}$, $P_{\Lambda=\Lambda_{IR}} = P_{gaussian}$,
 ϕ is a field living in infinite-dimensional function space; P_Λ is a functional on field space.

$\Lambda_{UV} \sim \text{unit scale}$, $\Lambda_{IR} \sim 1/L$.

Are diffusion model and RG isomorphic?

Diffusion model (data space)

Sample $x \in \mathbb{R}^d$ (e.g. image vector)

Marginal density $p_t(x)$

“Energy” / potential $U_t(x) \equiv -\log p_t(x)$

Score (what we learn): $s_t(x) = \nabla_x \log p_t(x) = -\nabla_x U_t(x)$

Time / noise level t

Key identification: $t \sim \log \frac{\Lambda_0}{\Lambda}$ (increase $t \Leftrightarrow$ lower Λ)

Exact RG (field space)

Field configuration $\phi(\mathbf{r}) \in \mathcal{F}$ (a function $\mathbb{R}^D \rightarrow \mathbb{R}^N$)

Probability functional $P_\Lambda[\phi]$

Effective action $S_\Lambda[\phi] \equiv -\log P_\Lambda[\phi]$ (+const)

Functional score: $\Sigma_\Lambda[\phi] = \frac{\delta}{\delta\phi} \log P_\Lambda[\phi] = -\frac{\delta S_\Lambda[\phi]}{\delta\phi}$

RG scale Λ (coarse-graining)

UV action ($\Lambda_0 = \text{UV cutoff}$)

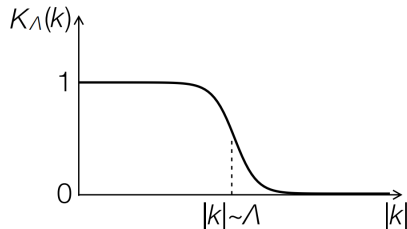
$$S_{\Lambda_0}[\phi] = \frac{1}{2} (\phi, G_0^{-1} \phi) + V[\phi], \quad Z_{\Lambda} := \int \mathcal{D}\phi e^{-S_{\Lambda}[\phi]}.$$

$G_0(k)$: Gaussian propagator, $V[\phi]$: interaction / non-Gaussian part, $(a, b) := \int_k a(k) b(-k)$.

Introduce a momentum cutoff $K_{\Lambda}(k)$

$$C_{\Lambda}(k) := K_{\Lambda}(k) G_0(k) \quad (\text{regulated propagator}),$$

$$K_{\Lambda}(k) = \frac{r(k^2/\Lambda^2)}{1 + r(k^2/\Lambda^2)}.$$



Scale-dependent action and physical invariance

$$S_{\Lambda}[\phi] = \frac{1}{2} (\phi, C_{\Lambda}^{-1} \phi) + V_{\Lambda}[\phi], \quad Z_{\Lambda} := \int \mathcal{D}\phi e^{-S_{\Lambda}[\phi]}.$$

Physical observables must not depend on where we place the cutoff. $\Rightarrow \partial_{\Lambda} Z_{\Lambda} = 0$

$$S_\Lambda[\phi] = \frac{1}{2}(\phi, C_\Lambda^{-1}\phi) + V_\Lambda[\phi], \quad C_\Lambda(k) = K_\Lambda(k) G_0(k).$$

Introduce the generating functional

$$Z_\Lambda[J] := \int \mathcal{D}\phi \exp\left(-S_\Lambda[\phi] + (J, \phi)\right).$$

$$\partial_\Lambda Z_\Lambda = 0 \quad \Longleftrightarrow \quad \partial_\Lambda Z_\Lambda[J] = 0 \quad \text{for all sources } J.$$

... (Omit derivation process)

Polchinski equation

$$\partial_\Lambda V_\Lambda[\phi] = \frac{1}{2} \int_k \partial_\Lambda C_\Lambda(k) \left[\frac{\delta V_\Lambda}{\delta \phi_k} \frac{\delta V_\Lambda}{\delta \phi_{-k}} - \frac{\delta^2 V_\Lambda}{\delta \phi_k \delta \phi_{-k}} \right]$$

Two terms (essential meaning)

- $(\delta V / \delta \phi)^2$: induced *tree* interaction from eliminating a thin momentum shell.
- $-\delta^2 V / \delta \phi \delta \phi$: *loop* fluctuation from shell self-contraction.

Polchinski equation

$$\partial_{\Lambda} V_{\Lambda}[\phi] = -\frac{1}{2} \int_k \frac{\partial_{\Lambda} K_{\Lambda}(k)}{k^2} \left(\frac{\delta^2 V_{\Lambda}}{\delta \phi_k \delta \phi_{-k}} - \frac{\delta V_{\Lambda}}{\delta \phi_k} \frac{\delta V_{\Lambda}}{\delta \phi_{-k}} \right). \quad p_{\Lambda}[\phi] \propto e^{-V_{\Lambda}[\phi]}.$$

V_{Λ} equation $\rightarrow p_t$ equation

$$\partial_t p_t(\phi) = -\frac{1}{2} \int_k \left[\frac{\partial_t K_t(k)}{k^2 K_t(k)} \frac{\delta^2 p_t}{\delta \phi_k \delta \phi_{-k}} + \frac{\delta}{\delta \phi_k} \left(\frac{\partial_t K_t(k)}{K_t(k)} \phi_k p_t \right) \right].$$

Here t is an RG time: $t = \log(\Lambda_0/\Lambda)$, $K_t(k) = K_{\Lambda(t)}(k)$.

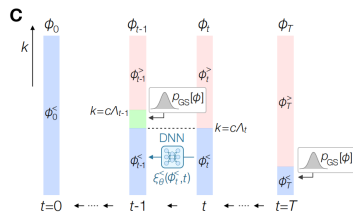
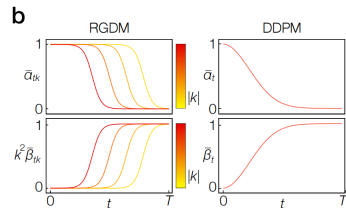
p_t equation mapped to a SDE

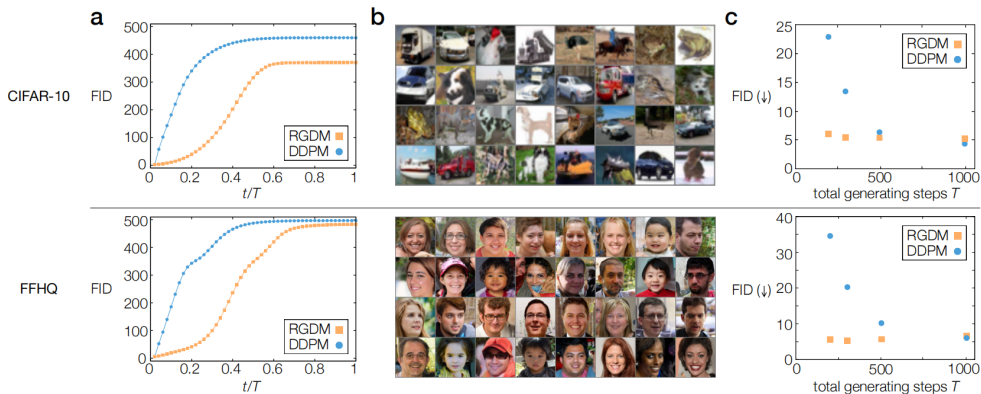
$$d\phi_{tk} = \frac{1}{2} \frac{\partial_t K_t(k)}{K_t(k)} \phi_{tk} dt + \sqrt{-G_0(k) \frac{\partial_t K_t(k)}{K_t(k)}} dw_k.$$

Discretize the SDE

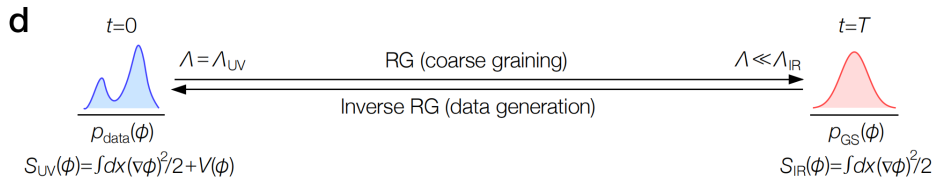
$$\phi_{tk} = \sqrt{\bar{\alpha}_{tk}} \phi_{0k} + \sqrt{\bar{\beta}_{tk}} \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, 1),$$

$$\sqrt{\bar{\alpha}_{tk}} = \exp\left(\frac{1}{2} \int_0^t ds \frac{\partial_s K_s(k)}{K_s(k)}\right) = \sqrt{K_t(k)}, \quad \bar{\beta}_{tk} = G_0(k)(1 - K_t(k)).$$





RGDM vs DDPM Image-generation experiments (CIFAR-10 / FFHQ)



DDPM

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

RGDM

$$\phi_{tk} = \sqrt{\bar{\alpha}_{tk}} \phi_{0k} + \sqrt{\bar{\beta}_{tk}} \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, 1),$$

$$\sqrt{\bar{\alpha}_{tk}} = \sqrt{K_t(k)},$$

$$\bar{\beta}_{tk} = G_0(k) (1 - K_t(k)).$$